

LESSON 124 (0.5) GROUND, (0.8) DUAL, and (0.8) SOLO

Lesson Objective

The purpose of this lesson is for the applicant to conduct their first solo flight. The flight will increase applicants' confidence and prepare them for future local and cross-country solo flights. Prior to the solo portion of this lesson, a dual flight is conducted to ensure the applicant is ready to solo under the current conditions.

Briefing

- Outline the objectives and completion standards of the lesson.
 - Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Review relevant airport diagrams with the student.
- Student questions.

References

- Airplane Flying Handbook. (FAA-H-8083-3C)
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
 - Chapter 14 - Airport Operations.

Instructor Guidance

- Do not endorse the student for solo until they are able to perform three landings without instructor input.
- Ideally, conduct this lesson at Melbourne (KMLB) to ensure sufficient lesson time and the availability of tower services to assist the student if needed during the solo portion.
- Borrow a handheld radio from the line department to monitor the student's radio communications and track their solo flight.
- During the dual portion, ensure at least one deliberate go-around is conducted. The student must demonstrate a correct and confident go-around procedure before soloing — hesitation on a go-around decision during solo operations is a significant safety risk.
- Brief the student on the specific airport diagram for the runways in use before departure. Ask them about expected taxiways, radio calls, runway incursion avoidance points, and hold short procedures. A student who arrives at the hold short line unsure of what to expect is not ready to solo.
- When the student is ready to solo, shut the engine down before exiting the aircraft. Brief the student one final time on the plan, remind them to announce "student pilot solo" on initial contact with Melbourne Ground, Melbourne Tower, and X59 CTAF.
- Complete the logbook endorsements accurately and in full compliance with 14 CFR §61.87 before the student departs for the solo. Double-check the endorsement wording, date, and your certificate number before signing.

Lesson Content (Ground)

Checklist Procedures

- Have the student run through all checklists.
- Incorporate flows for memory items.
- Student should be able to perform all briefings.
 - Setting correct flap configuration for line item in Before Takeoff Check
 - During Takeoff Briefing have student brief the type of takeoff being done

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Local Solo Brief

- Brief the plan of action to the student:
 - The first part of the flight will be a dual flight.
 - The student must demonstrate that they can conduct at least three safe takeoffs, traffic patterns, and landings.
 - No instructor guidance.
 - All landings must be full stop taxi backs.
 - Have the student take out their airport diagram.
 - Based on the runways in use, ask the student about what taxiways they should expect, radio calls, procedures, runway incursion avoidance, etc.
 - If the lesson is taking place at KMLB, the instructor will be getting dropped off back at the Florida Tech Aviation Ramp.
 - The engine needs to be shutdown prior to the instructor leaving the aircraft.
 - For guidance, refer to the a current KMLB airport diagram.
 - Advise tower than after the first 3 full stop taxi-backs it will be a student pilot solo, this will ensure that your spot in the pattern will not be taken.
 - The second part of the flight will be a solo flight.
 - As aforementioned, all landings must be full stop taxi backs.
 - The student should mention "student pilot solo" on initial contact with Melbourne ground/ Melbourne tower & X59 CTAF.
 - Monitor the student's radio calls through the portable radio.
 - Track the student's pattern visually if able or through an app (Flightradar24, FlightAware, etc.)

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FTA minimums.
 - Aircraft.
 - Airspace.

- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Ensure student is adhering to a Florida Tech Aviation standardized Checklist

Preflight Inspection

- Preflight is done with checklist in hand
- Identify fuel/oil levels, ensure required equipment & documents are onboard and overall aircraft is in an airworthy condition.

Engine Starting

- Correct engine start procedure is being implemented

Operation of the Communications and Navigation Units

- Demonstrate proper radio setup and frequency changes
- Verify navigation aids are tuned, identified, and operational

Radio Communication Procedures and ATC Phraseology

- Standard phraseology for clearance and ground/tower communication
- Emphasize listening skills and correct readbacks

Visual Scanning and Collision Avoidance

- Use FAA-recommended scan patterns
- Apply "see-and-avoid" with emphasis on traffic pattern operations

Taxiing

- Maintain centerline control with appropriate
- Use wind correction controls during taxi in varying conditions

Normal and Crosswind Takeoff

- Align and maintain runway centerline with rudder inputs
- Apply crosswind corrections on takeoff roll and initial climb.

Traffic Pattern

- The student will be in charge of all of the radio calls with tower and ground.
 - Emphasize the use of saying "say again" if the instructions are not clear.
 - Emphasize the use of saying "unable" if the instructions are not safe or cannot be followed.
- Traffic patterns are to be completed by the student without instructor guidance.

Go-Arounds or Rejected Landings

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted

Normal and Crosswind Landings

- Verify correct airspeeds.
- Verify timing of the round out and the flare.
- At least one main gear should be on the centerline.
- Longitudinal axis of the airplane must be aligned with the runway, no side-loaded landings

Normal and Crosswind Landings

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- Verify timing of the round out and the flare.
- At least one main gear should be on the centerline.
- Longitudinal axis of the airplane must be aligned with the runway, no side-loaded landings.

Power Off Landing

- While on the downwind, bring the power to idle.
- Ensure that the student pitches for 73 knots.
- Guide the student through when to turn base/final depending on the wind (if needed).
- The student must be able to land on the first third of the runway.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Lesson Content (Solo)

Checklist Procedures

- Student pilot solo

Engine Starting

- Student pilot solo

Operation of the Communications and Navigation Units

- Student pilot solo

Radio Communication Procedures and ATC Phraseology

- Student pilot solo

Visual Scanning and Collision Avoidance

- Student pilot solo

Taxiing

- Student pilot solo

Normal and Crosswind Takeoff

- Student pilot solo

Traffic Patterns

- Student pilot solo

Go-Arounds or Rejected Landings

- Student pilot solo

Normal and Crosswind Landings

- Student pilot solo

Postflight Procedures

- Student pilot solo.
- Ring the bell!
 - "Let it be known to all on this day, (day,month,year), Mr./Mrs xxx, has demonstrated the knowledge, skills, good judgement and mastery of the Piper Warrior to complete his/her first solo".

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 125.

Completion Standards

This lesson is complete when the applicant can:

1. Complete at least three stop and go or full stop normal and crosswind takeoffs and landings without instructor guidance while dual.

2. Execute a local solo in accordance with all Florida Tech Aviation, FAA and local procedures and regulations.

